Confidence-Aware Demand and Inventory Planning Tool Development Contract And Non-Disclosure Agreement

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1 Development Contract

1.1 Who this contract is between

This contract is between the originator, **Pratyush Tyagi**, who proposed this project, and the team of four students named below. The originator brings knowledge of how demand planners and inventory managers use forecasts. Each team member leads one part of the work and also writes one part of the final report.

Gil Benezer is the evaluation lead and the team's main contact. Gil measures how far off the forecasts are and how often the ranges hold the real sales, writes the fairness report, and builds the TimesNet model. Gil also sets up the check-in meetings and sends meeting notes to the originator.

Tarakaram Yadalla is the data lead. Tarakaram downloads the data and records its version, marks the sold-out hours, builds the daily data, and splits it for training and checking. Tarakaram also builds the DLinear fill-in model and the conformal prediction ranges, and deletes the raw data at the end of the project.

Jeena Shrestha is the dashboard lead. Jeena designs and builds the dashboard, including the product search, charts, tabs, and data editing. Jeena also builds the DLinear forecast model and leads the showcase demo.

Vishnu Purohitham is the compute and models lead. Vishnu sets up the computers, including Northeastern's Explorer cluster, and builds the TFT forecast model. Vishnu also runs the data security check and the final runs on the official test data.

1.2 What the team will build

The team will build a test version (a research prototype) of a planning dashboard for fresh-food stores. For each store and product, the dashboard shows past daily sales, the likely demand during hours when the product was sold out, and a demand forecast for the next 7 days. Each estimate comes with a range that shows how sure the system is. Demand planners and inventory managers

use the dashboard to choose which products to review and how much stock to keep. The system only gives forecasts and ranges; people make the decisions. The full details are in **Appendix A (the specification)**, which is part of this contract.

1.3 What the originator agrees to do

The project uses only public data (see Section A.3), so the originator does not need to share any data; if the originator later wants to share private data, it must first be covered by the NDA and agreed through the change process in Section 5 of this contract. The originator joins the check-in meetings (see Section 4 of this contract) and answers the team's questions, including questions about how planners and inventory managers work. Questions come by email or Microsoft Teams, and the originator answers within 3 days. The originator also looks at the work the team shares at each milestone (see Section 4 of this contract) and sends written feedback within one week. Finally, at the Week 13 showcase on December 8, 2026, the originator checks the system against the acceptance tests in Section A.7 and confirms in writing which tests passed.

1.4 What the team agrees to do

The team builds and hands over everything in Appendix A, on the dates in Section A.5, and passes the acceptance tests in Section A.7. The team meets the course milestones and shares the work with the originator at each one. The milestones in 2026 are the Interaction Walkthrough with storyboards of the AI features (October 20), the Midpoint Design Review (October 27), the pipeline skeleton (November 3), the ethics and impact milestone (November 10), the minimum working system (November 24), the dress rehearsal (December 1), the final showcase in Week 13 (December 8), and the final report and impact assessment (December 15). The team also meets the originator on Microsoft Teams every 2 weeks on Mondays, depending on progress, as agreed at the meeting on October 16, 2026. Gil sends short written notes after each meeting, and between meetings we use email or Teams chat. We tell the originator early if a date or goal is at risk. Finally, the team follows the data rules in Section A.3 and the NDA.

1.5 How to change the scope

Any change to the features, the data, the dates, or the acceptance tests needs a **written amendment** (a signed change note) that both sides sign. Small choices inside the agreed scope, like chart colors or the order of tabs, do not need an amendment; we record them in the meeting notes.

A change works in four steps. First, either side asks for the change in writing, by email or Teams, and says what should change, why, and what it affects. Second, we talk it over at the next check-in, or sooner if it is urgent; the team says how much time the change needs and what else may have to move or be dropped. Third, if both sides agree, the team writes the amendment with a number, the date, and the exact change to this contract or Appendix A. Fourth, the originator and all four team members sign the amendment (electronic signatures are fine). The change starts only when everyone has signed. Until then, this contract stays as it is.

If we cannot agree, this contract stays as it is, and either side may ask the course instructor for help. The team keeps every signed amendment with this contract.

1.6 How long this contract lasts

This contract starts when everyone has signed it. It ends when the team has handed in the final report (December 22, 2026) and deleted the raw data as set out in Section A.3.

1.7 Signatures

By signing below, each person agrees to this contract, including Appendix A. Electronic signatures are accepted.

Originator: Pratyush Tyagi

Signature: _____ Date: _____

Team member: Gil Benezer

Signature: _____ Date: _____

Team member: Jeena Shrestha

Signature: _____ Date: _____

Team member: Tarakaram Yadalla

Signature: _____ Date: _____

Team member: Vishnu Purohitham

Signature: _____ Date: _____

Appendix A: Development Contract Specification

A.1: Problem and Users

Retailers must make inventory decisions without knowing exactly how much customers will purchase. Many forecasting approaches give a single estimate without saying how dependable it is, so planners cannot recognize unreliable predictions, which can lead to shortages or excess inventory. The problem is harder when products are out of stock, because sales records miss the demand that could not be met, and products with limited sales history are especially vulnerable.

The primary users are demand planners and inventory managers. Demand planners examine historical and predicted demand for each store-product pair, review products flagged as uncertain, and correct estimates when needed; inventory managers use forecasts and their uncertainty to decide stock levels and replenishment. Indirect stakeholders are customers, suppliers, warehouse and store staff, and finance teams; we give particular attention to low-selling and niche products, which may otherwise receive less review. At the October 16 meeting, the originator confirmed this problem and these users but asked for a simpler interface: SKU search, daily demand charts, and straightforward uncertainty estimates, without model explanations (Section A.6).

A.2: AI features

At the October 16, 2026 meeting, the originator agreed to two AI features; the forecast (Feature 2) learns from the demand that Feature 1 recovers.

Feature 1: Recover the demand hidden by stockouts

Input	Hourly sales and hourly stock-out flags for each store-product pair in FreshRetailNet-50K, with discount, holiday, activity, and weather covariates.
Model	A recovery model (DLinear or TimesNet, following the dataset paper) with quantile outputs.
Output	An estimate of demand for every sold-out hour, with an uncertainty range, so the history shows what customers wanted and not only what was sold.
Who acts	The demand planner, who reads the recovered history to plan sales and business levers such as discounts and promotions.
When wrong	Estimates are always shown with their range, never as real sales. A planner who believes one is wrong edits the demand data directly; the system logs the edit and its reason and reruns both models on the corrected data. Accuracy is tested by hiding in-stock hours and comparing the estimate with what actually sold.

Feature 2: Forecast the next 7 days with an uncertainty range

Input	Daily demand history (observed plus recovered by Feature 1), past covariates, and planned future covariates such as discounts and holidays (uncertainty in future weather is a stretch goal; see Section A.6).
Model	A forecasting model (DLinear or TFT). Ranges come from quantile outputs calibrated with conformal prediction.
Output	A daily forecast for the next 7 days for each store-product pair, with 80% and 95% ranges. A product is flagged for review when its range is unusually wide or recent demand keeps landing outside it.
Who acts	The demand planner, who uses the forecast and its range to plan sales, and the inventory manager, who sets safety stock and reorder points from the range.

When wrong	Flagged products go to the planner; nothing is applied automatically. The planner corrects a forecast by editing the demand or covariate data behind it and rerunning, and every edit is logged with its reason. Calibration (PICP, ECE, reliability diagrams) is checked overall and for low- and high-selling products.

A.3: Data

Our main dataset is FreshRetailNet-50K, a public dataset on Hugging Face from Dingdong, a fresh-food online store in China: 50,000 store–product series (898 stores, 18 cities, 863 fresh products) from March to June 2024, with about 90 days to train on and 7 to test on per series. We use only these fields: number codes for city, store, product, and product group; date; daily and hourly sales; hourly sold-out flags; discount rate; holiday and activity flags; and daily weather (rain, temperature, humidity, wind). The originator provides no data, as there is no company data to share. The system creates estimated demand, forecasts with ranges, test scores, a run log (data version, code version, settings), and a planner edit log. If the main dataset is taken down, we switch to Favorita, M5, or Walmart data from Kaggle through the change process (contract, Section 5).

Under its CC BY 4.0 license, we may use, change, and show the data, including in class and at the showcase, if we credit the source and say what we changed. Because it is public, it is not confidential under the NDA; Kaggle backups follow each competition's rules. We add no outside data that could reveal a store, city, product, or supplier, and do not try to find out what the number codes stand for. Only the four team members and course staff can access the data, kept in a private team folder on team laptops and Northeastern's Explorer cluster and never in a public code repository. Planner edits (Section A.2) go to a separate edit log (store, product, day, old and new value, date, reason), so the original dataset never changes; this log is new since the proposal, which only recorded overrides. We record the download date and version, since the dataset could change or disappear. We keep raw data, model files, and logs until the final report (December 15, 2026); by December 22, 2026, each member deletes their copies, leaving only code and summary results, and Tarakaram (data lead) confirms this to the originator in writing.

Known gaps: the data has no delivery or lead times, stock on hand, safety stock, reorder points, orders, prices, or costs (only a discount rate), so we cannot forecast delivery delays, chart real stock, or show money saved (Section A.6). Sales are scaled by an unknown number, so forecasts are not in real units; products have codes but no names, so search works by code; and weather is the recorded daily value, not a forecast, while store locations are hidden, so live weather cannot be added. Low-selling products have few sales, many zeros, and more error (about 37%, vs. 23–28% for high sellers, in the dataset paper), so we report them separately. The data covers only three months from one retailer in China with no launch dates, so yearly patterns and new products cannot be studied and results may not carry over to other stores; how the sold-out flags were checked is not published.

A.4: Evaluation

For accuracy, we use Weighted Absolute Percentage Error (WAPE) and Mean Absolute Scaled Error (MASE) for error size, and Weighted Percentage Error (WPE) for bias direction, where A_t is actual and F_t forecast demand, h is the number of forecast days, and n the number of training

days:

$$\text{WAPE} = 100\% \cdot \frac{\sum_t |A_t - F_t|}{\sum_t A_t}, \quad \text{WPE} = 100\% \cdot \frac{\sum_t (A_t - F_t)}{\sum_t A_t}, \quad \text{MASE} = \frac{\frac{1}{h}\sum_{j=1}^{h} |A_j - F_j|}{\frac{1}{n-1}\sum_{t=2}^{n} |A_t - A_{t-1}|} \quad (1)$$

For calibration, we use prediction interval coverage percent (PICP), the share of true values inside the 95% interval formed by the 2.5% and 97.5% quantiles (trained with quantile, or pinball, losses), so good calibration means PICP ≈ 0.95. Since a wide enough interval reaches any coverage, we also report mean prediction interval width (MPIW); at a given PICP, lower MPIW is better:

$$\text{PICP} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\big(\hat{q}_i^{2.5\%} \le y_i \le \hat{q}_i^{97.5\%}\big), \qquad \text{MPIW} = \frac{1}{N}\sum_{i=1}^{N} \big(\hat{q}_i^{97.5\%} - \hat{q}_i^{2.5\%}\big) \quad (2)$$

All models undergo at least 3-fold cross-validation. If the official test split is the 7 days after training for every series (to be confirmed), imputation models are scored on held-out training data and forecasting models on the test days, to mirror intended use.

A.5: Deliverables and Timeline

Week	Course milestone	Planned deliverables
6 (Oct. 20)	Interaction Walkthrough	Dashboard storyboard and user interaction flow; initial data processing and evaluation code.
7 (Oct. 27)	Architecture & Tooling Seminar; Midpoint Design Review	Final system architecture and shared output format; start of the imputation and forecasting models.
8 (Nov. 3)	Pipeline skeleton	Complete pipeline from data preparation through imputation and forecasting to a basic dashboard.
9 (Nov. 10)	Ethics and impact milestone	Evaluation of accuracy, uncertainty, and fairness across product groups.
10 (Nov. 17)	Deployment & Monitoring	Confidence labels and demand-change warnings; monitoring and retraining plan.
11 (Nov. 24)	Cybersecurity for AI Systems (Minimum Working System)	Minimum working system with model comparisons, a product review queue, and security checks.
12 (Dec. 1)	Continual Learning & Dress Rehearsal	Dashboard refinements, final evaluations, and a practice demonstration.
13 (Dec. 8)	Final Showcase	Working prototype demo: forecasts, uncertainty estimates, dashboard, and evaluation results.
14 (Dec. 15)	Final Report & Impact Assessment	Final report, evaluation results, source code, and documentation to reproduce the prototype.

Final handover: by December 15, 2026, the team delivers the research prototype, source code, evaluation results, and technical documentation to the originator.

A.6: Out of Scope

As the brief requires (Section A.8) and the meeting confirmed, the system is a research test version that the team runs offline during the semester: no link to live ordering, ERP, or store systems, no real orders, and no real users after the course. It will not forecast delivery delays or lead times, since no public dataset we know of has delivery dates; the originator's past work did this, but both sides agreed at the meeting that it is not possible here. It will not make stock decisions (no automatic reordering; no safety stock, reorder points, minimum stock, or order amounts; no

chart of real stock levels), because the data has no stock or order records and people set stock levels from the ranges. This is a change: the originator asked for a stock chart, and the proposal (Section 5) mentioned safety-stock checks and a minimum stock level for niche products.

The dashboard will not show explanation scores (such as SHAP values), "main reasons", per-product past coverage, or low-data flags, because the originator asked us to keep to the uncertainty range (a change from the proposal, Section 5). It will not show hourly charts, since the originator asked for daily views; the models still use the hourly data. The model is not retrained right after each edit; edits are used at the next forecast run. We will not use live weather, private company data, or anything that could reveal a store, and will not make pricing decisions, rank stores or staff, or try to find out what the number codes stand for (proposal's governance plan).

Stretch goals, done only if time allows and not checked at the showcase, are weather uncertainty (small random changes to future weather values, to see how much the 7-day range widens, as the originator asked), a text summary next to each chart, test-time augmentation and an external interval network (after conformal prediction), and sold-out prediction. Anything not named in Section A.2 is out of scope unless added through the change process (contract, Section 5).

A.7: Acceptance Criteria

1. Four separate models, two for imputation and two for forecasting, are evaluated on the metrics in Section A.4.
2. All models perform within one standard deviation of published WAPE values, based on the 3-fold (or more) cross-validation.
3. All models have a mean PICP above 80%.
4. The models selected for deployment have the lowest MPIW, as uncertainty quantification is the priority of the project.
5. A web interface for the best-performing model has an SKU search bar that shows a product or store-product time series with uncertainty estimates, and tabs including (a) a chart of historical and future demand with 95% intervals, (b) a table for manually overriding the underlying data, and (c) a chart of covariate time series.

A.8: Scoping Conditions

No written scoping conditions were set at selection (proposal, Section 5), so we adopt the constraints in the originator's project brief as terms of this contract, copied word for word:

(a) *"Public/synthetic retail and demand datasets only (e.g. Kaggle retail sales, M5-style competition data)".* Our data source under this term is FreshRetailNet-50K (Dingdong-Inc, Hugging Face, CC BY 4.0).
(b) *"no proprietary Navanta or client data."*
(c) *"Not building a live production planning system or integrating with a real ERP; the deliverable is a research prototype demonstrating confidence-aware forecasting and human-in-the-loop escalation."*
(d) *"One academic term, so scope is a bounded product catalog and one or two demand patterns done rigorously rather than full multi-echelon supply-chain coverage."*

2 Non-Disclosure Agreement (NDA)

Project: Confidence-Aware Demand and Inventory Planning Tool
Course: CS6170 – AI Capstone, Fall 2026
Effective Date: 2026-10-19

Parties

This agreement is between Pratyush Tyagi (the Originator) and the project team, Gil Benezer, Jeena Shrestha, Tarakaram Yadalla, and Vishnu Purohitham (the Team), students in CS6170 AI Capstone at Northeastern University, Fall 2026. It covers the project A Confidence-Aware Demand & Inventory Planning Assistant. It starts when the last party signs, and its duties last until December 31, 2027, unless the Originator releases them earlier in writing.

2.1 What is confidential

At the meeting, the Originator was not sure whether anything needs protection. This agreement therefore names the material most likely to matter and lets him add more later. The following is confidential:

(a) The Originator's descriptions of his prior industry work, including the models he trained to estimate delivery delay times and size safety stock, and how those models were built, tuned, and evaluated.

(b) The specific pre-trained models, settings, or sources the Originator recommends from that work, where he says they are not public.

(c) Internal details of his previous employer's processes, clients, data, thresholds, or results that he describes. The general methods named in the project brief (ABC-XYZ classification, champion-challenger model selection, and safety stock and reorder point sizing) are not confidential.

(d) Any business plans, product ideas, or startup plans the Originator shares about this project.

(e) Meeting notes, recordings, and messages, to the extent they contain items (a) to (d).

2.2 What is not confidential

(a) The FreshRetailNet-50K dataset and all results computed from it, which are public under CC BY 4.0.

(b) Published papers, public documentation, and anything that is or becomes public through no fault of the Team.

(c) Anything a team member already knew, or developed independently without using confidential material.

(d) The Team's code, models, dashboard, storyboards, and reports, as long as they contain no confidential material.

(e) Anything the university or the law requires to be disclosed. Where allowed, the Team tells the Originator first.

2.3 3. Who may access it, and where it is stored

(a) Confidential information will be accessible only to the development team and authorized course staff when necessary for academic evaluation.

(b) All confidential materials will be stored in a private, access-controlled team folder and will not be shared with unauthorized individuals or uploaded to public repositories.

2.4 Public Presentations

(a) The team may present the project design, system architecture, dashboard, and evaluation results using public, anonymized data during class presentations and the Week 13 Final Showcase.

(b) Any confidential information provided by the originator will not be disclosed publicly without prior written permission.

2.5 End-of-Project Handling

(a) Only the four team members may access confidential material. The instructor and teaching assistants may see it only when needed to grade or advise, and the Team tells them it is confidential.

(b) It is stored only in the team's private shared folder on a Northeastern-managed account and in the team's private GitHub repository, both limited to team members. It is never placed in a public repository, a public channel, or a personal device outside those folders.

(c) It is not pasted into AI tools or other services that keep or train on what they receive, unless the Originator agrees in writing.

2.6 Signatures

By signing below, each person agrees to this contract, including Appendix A. Electronic signatures are accepted.

Originator: Pratyush Tyagi

Signature: _____ Date: _____

Team member: Gil Benezer

Signature: _____ Date: _____

Team member: Jeena Shrestha

Signature: _____ Date: _____

Team member: Tarakaram Yadalla

Signature: _____ Date: _____

Team member: Vishnu Purohitham

Signature: _____ Date: _____

3 Meeting Notes

3.1 Meeting Summary of Six Questions

1. What is the problem and who are the people who will act on the system's output?
   (a) The problem is that current demand forecasting methods are inaccurate and do not yield calibrated uncertainty estimates.
   (b) The people who will act on the system outputs are the demand planners and inventory stockers for a hypothetical grocery e-commerce business.
2. What are the AI features of the system?
   (a) Feature 1: Demand Imputation Model
      i. A neural forecasting model that imputes historical store-product demand during stockout events and/or other censoring instances using both prior data and covariates.
      ii. This will be used by demand planners to assist with planning sales and other manipulation of covariates under business control
   (b) Feature 2: Demand Forecasting Model
      i. A neural forecasting model that uses imputed and actual historical demand, along with covariates, to forecast store-product demand over a 7-day period.
      ii. This will be used by both demand planners for the prior activities listed as well as by inventory stockers to evaluate and inform safety stock ordering.
3. What data will our team access to develop the product, under what terms, and for how long?
   (a) We plan to use the public FreshRetailNet-50k (`https://huggingface.co/datasets/Dingdong-Inc/FreshRetailNet-50K`) dataset through HuggingFace.
   (b) The terms of data access are governed by the Creative Commons Attribution 4.0 International License (CC BY 4.0) available at `https://creativecommons.org/licenses/by/4.0/legalcode`.
   (c) We plan to use the dataset until the end of the semester.
4. What does "done" look like at Week 13?
   (a) An evaluation of two different combinations of models we will re-implement from the dataset paper (`https://arxiv.org/abs/2505.16319`; DLinear-DLinear and TimesNet-TFT) along both forecast accuracy and uncertainty calibration dimensions.
      i. The model combinations will be trained to estimate the 2.5% percentile and 97.5% percentile using the respective pinball/quantile loss functions, yielding both a 95% confidence interval and the demand prediction derived from the midpoint.
      ii. Both model combinations will be evaluated on at least 3 splits of provided test data, and the mean of the test Weighted Absolute Percent Error (WAPE, AKA MAPE) values must be in 1 standard deviation of the published results for WAPE.
      iii. The model combinations will also be evaluated based on the mean absolute scaled error (MASE, `https://en.wikipedia.org/wiki/Mean_absolute_scaled_error`) as suggested by the client.

iv. The models will be conformalized to ensure that the estimated intervals have statistical guarantees (according to prior work).

v. The models will be evaluated with respect to calibration and sharpness based on prediction interval coverage percent (PICP) and mean prediction interval width (MPIW) respectively.

vi. The most accurate model with the most well calibrated, sharpest estimate of the 95% confidence interval will be moved forward to the next output.

vii. Some of the anticipated libraries for forecasting and uncertainty estimation to be used are neuralforecast (`https://github.com/Nixtla/neuralforecast`), TorchUncertainty (`https://torch-uncertainty.github.io/index.html`), and PytorchForecasting (`https://pytorch-forecasting.readthedocs.io/en/stable/index.html`)

(b) A data visualization dashboard to show, for the FreshRetailNet-50k dataset, time series graphics for a store-product combination(s).

i. The web interface will be separated by tabs to ensure user-friendliness.

ii. There will be a search bar to enable the lookup for a particular SKU.

iii. 95% confidence intervals and midpoints will be shown both for historic stockout periods as well as for 7 day forecast periods.

iv. There must be an interface for a user to be able to enter actual demand/stock as a manual override that changes underlying data.

v. In theory, this new data would be fed back into the model and after a certain accumulation of new human-entered data, the models would be retrained.

5. What is out-of-scope?

(a) The output is a research prototype and not a live system.

(b) Developing the application for actual streaming data is out of scope.

(c) Integration with a live ERP is out of scope.

6. How often will the client keep in contact with the team?

(a) Every 2 weeks on Mondays.